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Abstract

Perineural invasion (PNI) is a pivotal prognostic indicator in pancreatic cancer and influences risk stratification and treatment decisions. However, conventional computational pathology methods often rely on random patch sampling and may fail to capture the complex tumor–nerve topology and hierarchical spatial organization of the tissue microenvironment. We propose NCHG, a **Nerve-Centric Heterogeneous Graph framework** that explicitly models nerve-oriented invasion patterns for survival prediction. NCHG constructs a hierarchical tissue-level graph centered on neural structures and integrates three node types: nerve, tumor, and microenvironment nodes. To better characterize neural involvement, we introduce **intra-neural tumor nodes** based on their geometric location within the reconstructed nerve and proximity to the nerve boundary. A biologically inspired **dual-attention mechanism**, combining Structural Attention and Feature Similarity Attention, captures spatially directional tumor–nerve interactions and infiltration patterns. NCHG achieves a C-index of 0.6437 and an AUC of 0.6857 on TCGA-PAAD, and 0.6504 and 0.7041 on GL, respectively. On the independent SZY cohort, it achieves a C-index of 0.6243 and an AUC of 0.6625, demonstrating superior performance and strong cross-center generalization.

Methods

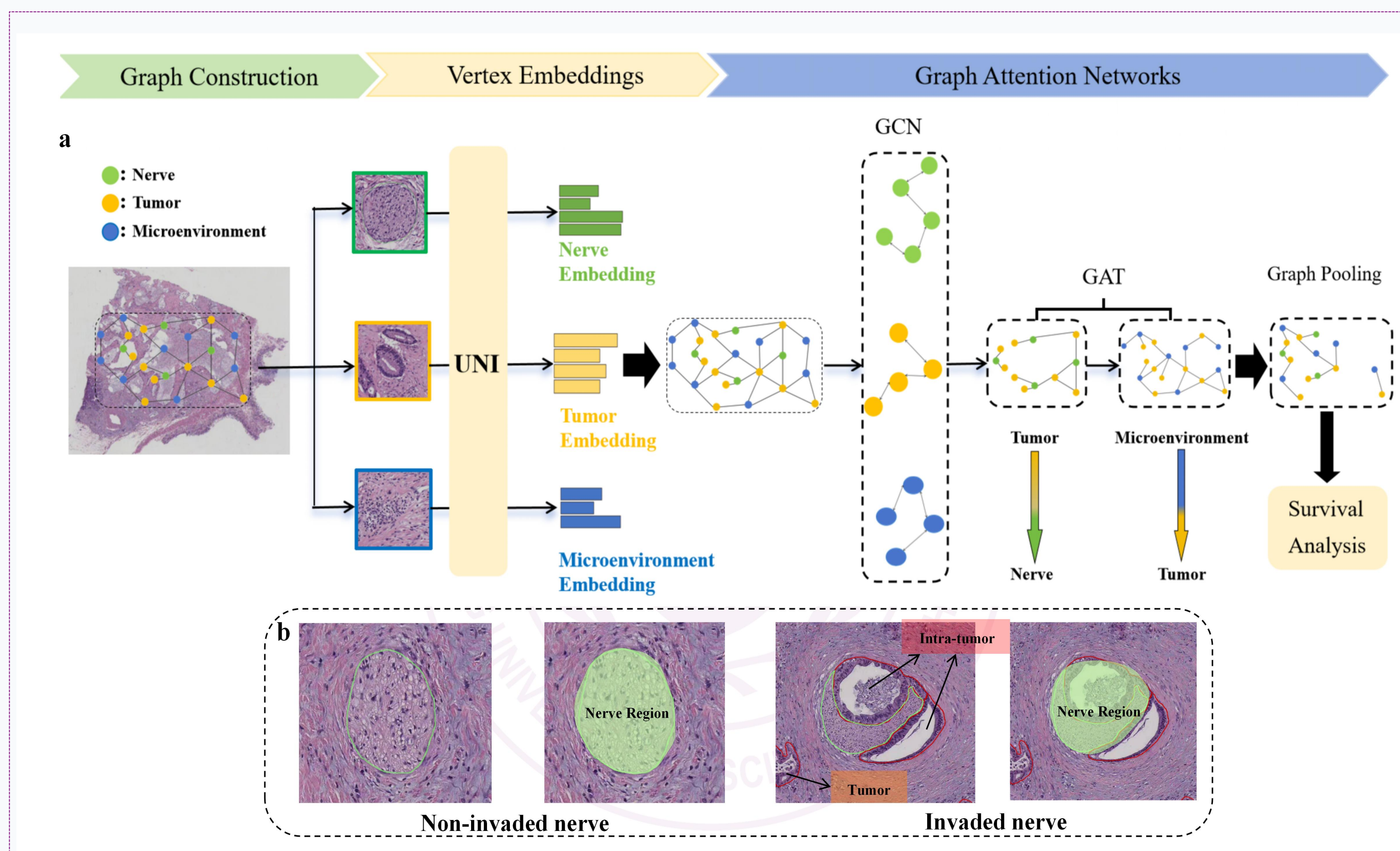


Figure. 1. Overview of the NCHG framework. (a) Construction of a heterogeneous spatial graph across nerve, tumor, and microenvironment regions, with node features initialized by UNI embeddings. A hybrid GCN-GAT architecture with graph pooling models hierarchical interactions, and the resulting representations are used for survival prediction. (b) Illustration of non-invaded and invaded nerves.

Key Points

- We reconstruct 2D nerve geometry via polygon fitting and convex hull, partitioning tissue into the reconstructed nerve, boundary-adjacent perineural microenvironment (168 μm), and distal regions (168/392/560 μm). A tumor node inside the nerve and boundary-adjacent area is defined as **intra-neural tumor nodes** capturing tumor infiltration and boundary attachment as the node-level basis for NCHG.
- We design a **dual-attention** mechanism capturing both feature similarity and spatial topology. **Structural Attention** uses Gaussian distance decay with adaptive bandwidth, while **Feature Similarity Attention** captures phenotypic consistency. By fusing both attentions, the model enhances intra-neural tumor nodes while modeling the nerve and its microenvironment, thereby improving survival prediction.

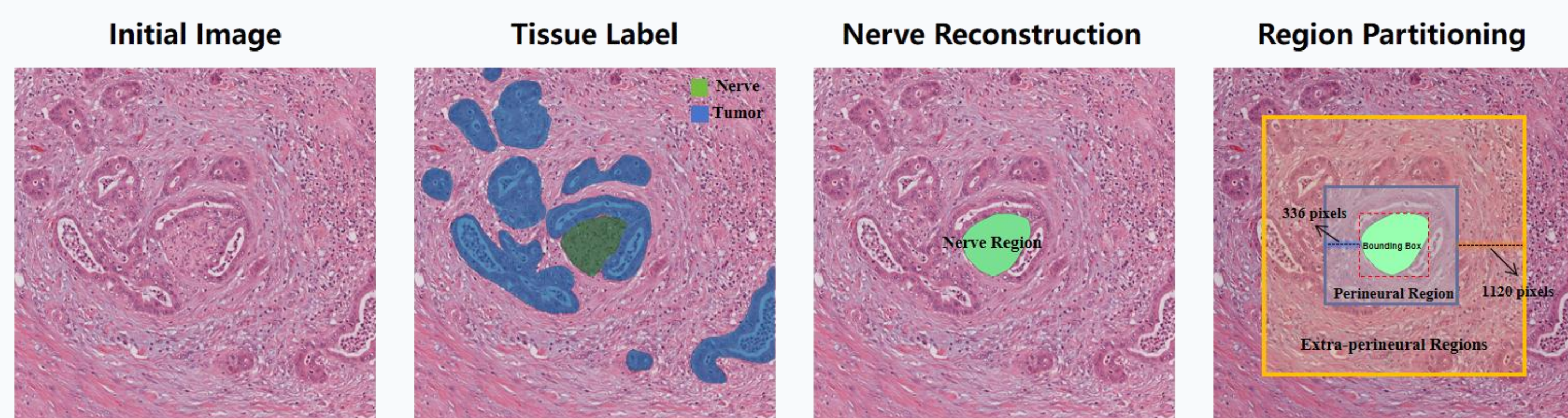


Figure. 2. Nerve-Centric Tissue Partitioning and Definition of Intra-Neural Tumor Nodes.

Conclusion

- NCHG is a nerve-centric heterogeneous graph framework that models tumor–nerve interactions for Pancreatic Ductal Adenocarcinoma survival prediction by defining intra-neural tumor nodes and applying dual-attention message passing, showing strong performance and interpretability across three cohorts. Future work will improve robustness to segmentation errors via more reliable models and end-to-end learning, integrate multiomics data, extend NCHG to other PNI-related cancers, validate it in larger prospective multi-center cohorts, and translate it into a practical tool for PNI assessment in precision oncology.

Results

- A total of 386 PAAD patients from three independent cohorts were included: TCGA-PAAD (n = 186; 90 censored/96 uncensored), GL (n = 146; 83/63), and SZY (n = 54; 12/42). TCGA-PAAD and GL were used for model development and internal evaluation, while SZY served as an external validation cohort to assess generalizability.

Table 1. Quantitative comparison of survival prediction performance on TCGA-PAAD and GL datasets.

Method	TCGA-PAAD		GL	
	C-index	AUC	C-index	AUC
WSISA [20]	0.5521 $\pm$ 0.04	0.5942 $\pm$ 0.03	0.5642 $\pm$ 0.05	0.6159 $\pm$ 0.05
DeepAttnMISL [21]	0.5864 $\pm$ 0.03	0.6742 $\pm$ 0.04	0.5354 $\pm$ 0.05	0.6216 $\pm$ 0.03
Patch-GCN [9]	0.6067 $\pm$ 0.02	0.6641 $\pm$ 0.04	0.6346 $\pm$ 0.05	0.6835 $\pm$ 0.03
DeepGraphSurv [22]	0.5574 $\pm$ 0.03	0.6050 $\pm$ 0.06	0.5346 $\pm$ 0.05	0.5860 $\pm$ 0.04
HEAT [23]	0.6342 $\pm$ 0.05	0.6941$\pm$0.03	0.6554$\pm$0.02	0.6842 $\pm$ 0.04
NCHG (ours)	0.6437$\pm$0.05	0.6857 $\pm$ 0.04	0.6504 $\pm$ 0.04	0.7041$\pm$0.05

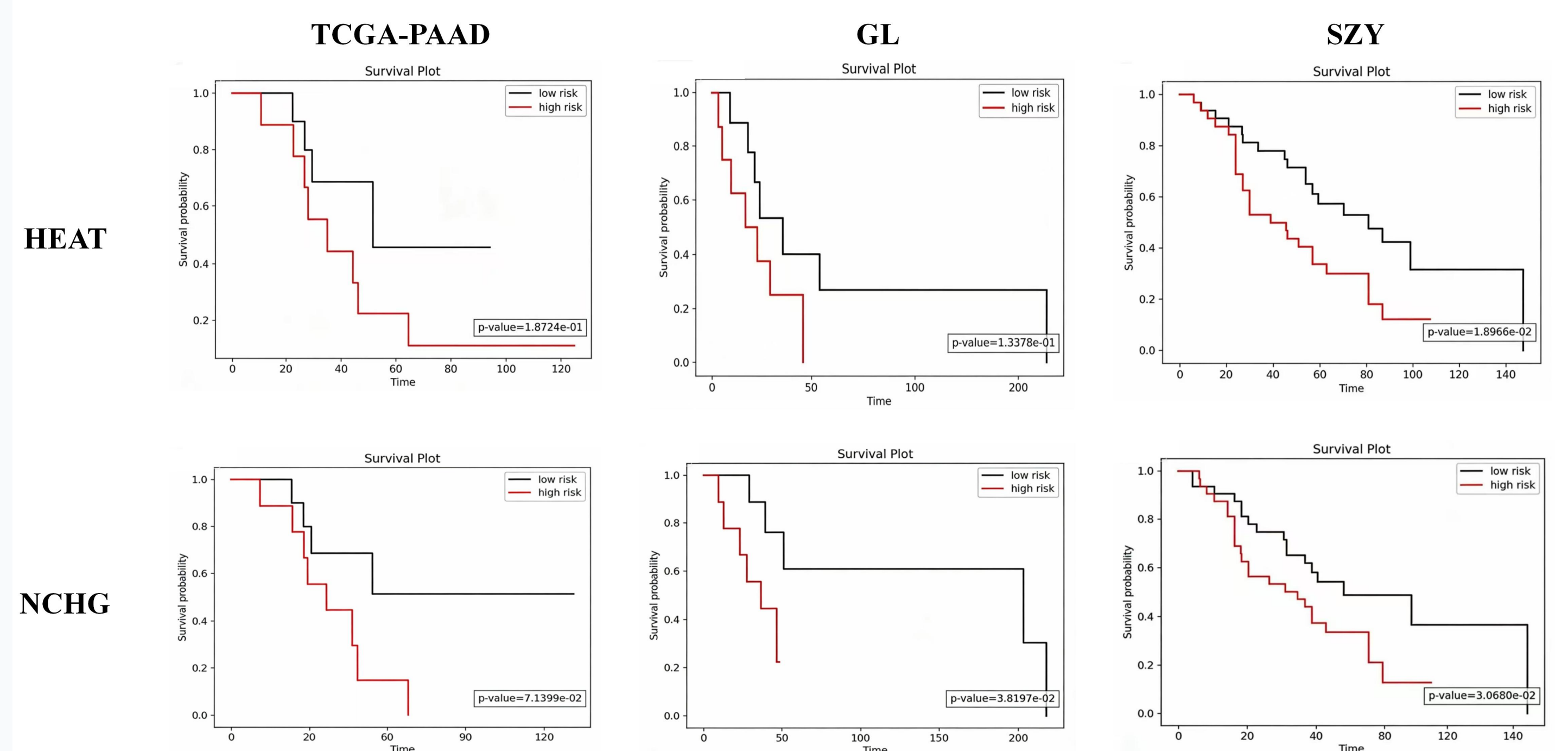


Figure. 3. Comparison of the Stratification Performance Among Different Methods.

Interpretability

- Attention maps (Figure. 3) show NCHG focuses on invaded nerves; poor prognosis attends to intra-neural tumor infiltration, while long-term survivors attend to the perineural microenvironment, highlighting neural topology as a survival determinant.

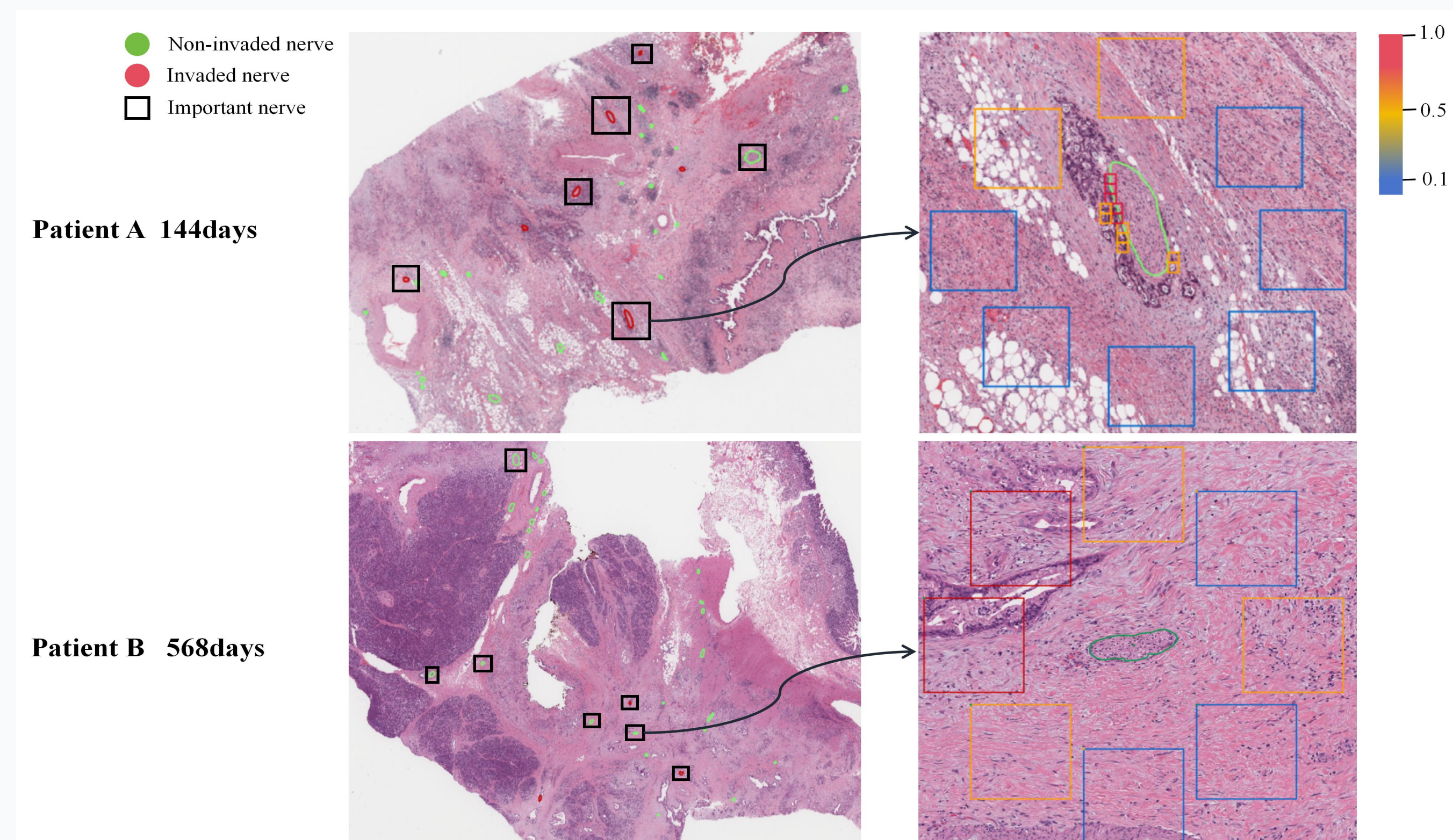


Figure. 4. Attention maps of neural involvement across prognostic groups.